Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Personalized Networking Assistant codebase in terms of code organization, readability, and reusability. The project follows a modular architecture using Python, FastAPI, Streamlit, HTML, CSS, SQLite, DistilBERT, GPT-2, and the Wikipedia API, making the application easy to understand, maintain, and extend.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code	Yes	Python PEP-8 style followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, and application
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names used
4	Function / Method Names	Functions use meaningful and descriptive names	Yes	Functions describe their operations clearly
5	Code Comments	Comments explain important logic and functionality	Partial	Comments added for major sections only
6	Modular Design	Code is divided into reusable modules	Yes	Authentication, recommendation engine, AI models, and database modules are separated
7	No Redundant Code	Duplicate or unnecessary code is removed	Yes	Authentication, recommendation engine, AI models, and database modules are separated
8	Error Handling	Exceptions and validation are implemented	Yes	Input validation and exception handling are implemented using FastAPI

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, FastAPI	Handles routing and prediction requests	High
2	User Authentication Module	Python	Login and Registration	High
3	Recommendation Engine	Python, DistilBERT	Generates personalized networking recommendations	High
4	Conversation Generator	Python, GPT-2	Generates AI conversation starters	High
5	Fact Verification Module	Wikipedia API	Verifies networking information	High
6	Database Module	SQLite	Stores user profiles, conversation history, and feedback	High
7	Streamlit User Interface	Streamlit, HTML, CSS	Dashboard, Profile, Recommendation Pages	High
8	Utility Functions	Python	Input validation, preprocessing, and helper functions	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder structure
Readability	5	Meaningful naming conventions and simple logic improve readability
Reusability	5	Modular design enables code reuse across the application
Documentation / Comments	4	Major sections are documented; additional comments can further improve clarity
Overall Score	4.95	The project demonstrates good coding practices and maintainable design suitable for future enhancements.